Documentation

A guide to using LaTeX File Template v.3.0.0.

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Contents

1 Introduction

Thank you for choosing to use LaTeX File Template. If you find any bugs or want some feature not currently included, then my contact information can be found on my GitHub: https://github.com/MatthewEmer.

This document is intended for academics (with special focus on those in the fields of computer science or mathematics) who want to write papers or other related documents in a consistent format which ties in with standard practises from other published papers. This guide consists of setup instructions to download the files and create your first document, before moving onto the custom commands/environments designed to speed up academic writing. Before using this guide, I would suggest that you have some practise with some basic LaTeX commands using a guide such as https://www.overleaf.com/learn.

Accurate as of Version 3.0.0.

2 Document Setup

2.1 The Files

The actual LaTeX file template is contained in the folder *The Template* and consists of four files: *template.tex*, *_documentContent.tex*, *_appendix.tex*, and *template.pdf*. To use the template, all four files must be downloaded and placed in the same folder.

2.1.1 template.tex

This file sets up the structure of the document, as well as acting as a compiler for the other two *.tex* files, ensuring they get inserted into the right sections of the document. When creating your document, you need to edit lines 12-15 of this file to create the title page, alongside the document header and footer, as described in Table 1.

Line	Variable Name	Description
12	thetitle	The document's title. This will be placed in the header.
13	thesubtitle	The document's (optional) subtitle. This will be placed in the header.
14	theauthor	The list of authors. This will be placed in the footer.
15	thedate	Replace the code if you want a static date rather than date of last compile.

Table 1: Lines to edit.

2.1.2 _documentContent.tex

This file allows you to add all the main content of the document you are trying to write. It will be inserted after the table of contents, but before the appendix (if you choose to add one). By default it contains an example section and definition. Only pages in this file are counted towards the page count of the document.

2.1.3 _appendix.tex

This file allows you to add an optional appendix to the end of your document which is not counted towards your page count. By default it contains a unnumbered part title which has been inserted into the table of contents.

2.1.4 template.pdf

This file contains the output of the LaTeX compiler. If you want to rename this document while still working on it, then delete this version, alongside any build files your compiler has created and change the name of *template.tex*.

2.2 Additional Notes

To add images to the document, create a subfolder called *Images* within the same folder as the template and store any images you wish to use in there.

If placing the document within a GitHub repository, you may want to copy over the *.gitignore* from this repository as it is set up to ignore any LaTeX build files.

3 Custom Commands

3.1 Academic Environments

adding some nonsense

3.2 Mathematics

3.2.1 Set Notation

Command: $\backslash R$, $\backslash N$, $\backslash C$, $\backslash Q$, $\backslash Z$, $\backslash I$

Inserts the symbol for the given number set.

No.	Name	Description	Example Value
n/a			

Table 2: Parameters for set notation.

Example Call: $\backslash R \quad \rightarrow \quad \mathbb{R}$

3.2.2 Matrix/Vector Operations

Command: $\backslash dotproduct$

Inserts the notation for a dot product operation between two given vectors.

No.	Name	Description	Example Value
1	Vector 1	The vector on the left of the dot product.	u
2	Vector 2	The vector on the right of the dot product.	v

Table 3: Parameters for dot product.

Example Call: $\backslash dotproduct\{u\}\{v\} \quad \rightarrow \quad \langle \mathbf{u}, \mathbf{v} \rangle$

Command: $\backslash hessian$

Inserts the notation for the Hessian of a given vector.

No.	Name	Description	Example Value
1	Vector	The vector that the Hessian is calculated from.	v

Table 4: Parameters for Hessian.

Example Call: $\backslash hessian\{v\} \quad \rightarrow \quad \mathbf{H}_v$

Command: $\backslash jacobian$

Inserts the notation for the Jacobian of a given vector.

No.	Name	Description	Example Value
1	Vector	The vector that the Jacobian is calculated from.	v

Table 5: Parameters for Jacobian.

Example Call: $\backslash jacobian\{v\} \quad \rightarrow \quad \mathbf{J}_v$

Command: $\backslash lagrange$

Inserts the notation for a Lagrangian.

No.	Name	Description	Example Value
n/a			

Table 6: Parameters for Lagrangian.

Example Call: $\backslash lagrange \quad \rightarrow \quad \mathcal{L}$

3.3 Computer Science

3.3.1 Join Operations

Command: $\backslash insertinnerjoin$, $\backslash insertleftouterjoin$, $\backslash insertrightouterjoin$, $\backslash insertfullouterjoin$

Inserts the relevant notation for the chosen join operation.

No.	Name	Description	Example Value
n/a			

Table 7: Parameters for join notation.

Example Call: $\backslash insertleftouterjoin \quad \rightarrow \quad ⟕$

3.3.2 Algorithms

Command: $\backslash insertalgorithm$

Inserts an algorithm block for pseudocode.

No.	Name	Description	Example Value
1	Algorithm name	The name of the algorithm being inserted.	Printing Hello World.
2	Input(s)	The input(s) required by the algorithm.	n/a
3	Output(s)	The output(s) returned by the algorithm.	n/a
4	Pseduocode	The actual pseudocode for the algorithm.	See below

Table 8: Parameters for inserting algorithms.

Example Call: $\backslash insertalgorithm$\{Printing Hello World.\}\{n/a\}\{n/a\}\{$code$\}:

Algorithm 1 Printing Hello World.

Input: n/a
Output: n/a

```
FUNCTION printHelloWorld()
  PRINT "Hello World"
ENDFUNCTION
```

3.4 Additional Commands

3.4.1 Text Formatting

Command: *colouredText*

Changes the colour of a group of text to a given colour.

No.	Name	Description	Example Value
1	Colour	The colour to change the text to.	red
2	Text	The text you want outputted in colour.	Hello World

Table 9: Parameters for coloured text.

Command: *separate*

Adds additional vertical spacing after the last element and removes the new paragraph indent before the next element.

No.	Name	Description	Example Value
1	Spacing	The number of points of vertical spacing.	5

Table 10: Parameters for separating.

Command: *comment*

Colours the given text red so its obvious that there is a comment in the document.

No.	Name	Description	Example Value
1	Text	The text to be written as a comment.	Hello World

Table 11: Parameters for comments.

3.4.2 Tables and Figures

3.4.3 Date Formats

Command: *monthyeardate*

When placed alongside a date, this command reformats it and outputs the full name of the month, followed by the four-digit year, as seen in the document title page.

No.	Name	Description	Example Value
n/a			

Table 12: Parameters for MMMM/YYYY dates.

Command: *yeardate*

When placed alongside a date, this command reformats it and outputs only the four-digit year.

No.	Name	Description	Example Value
n/a			

Table 13: Parameters for YYYY dates.